

Customer Churn & Retention Intelligence

Where churn concentrates in a 100,000-subscriber base, what it costs, and which customers to contact first.



OBSERVED RETENTION CURVE · 100,000 SUBSCRIBERS · FOUR YEARS FROM SIGNUP

32.68%

CHURN RATE

\$433,107

MONTHLY REVENUE LOST

100,000

SUBSCRIBERS ANALYSED

PREPARED BY

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OBSERVATION WINDOW

Jan 2019 – Dec 2025

METHOD

SQL · Python · pandas

01 Executive Summary

Nearly one in three subscribers has cancelled, removing **\$433,107 in monthly charges** — 28.98% of the book's total. That loss is not spread evenly across the base. It concentrates in an identifiable profile, and it is preceded by an event the business already records.

32.68%

CHURN RATE · 32,676 CUSTOMERS

67.32%

RETENTION RATE

28.98%

OF MONTHLY CHARGES LOST TO CHURN

\$1.49M

TOTAL MONTHLY CHARGES

\$14.95

AVERAGE REVENUE PER USER

642

AVERAGE TENURE (DAYS)

Four findings drive this report

1. Contract structure is the largest single point of separation.

Monthly-contract subscribers churn at **50.80%** against **20.10%** on annual terms. Monthly contracts hold 38.4% of total monthly charges but account for **62.1% of everything lost** — \$269,154 of \$433,107. Contract terms are set by the business at the point of sale, which makes this the most directly actionable lever available — subject to the causal caveat set out in Section 06.

2. Risk factors compound, and contract outweighs plan.

Basic on a monthly contract churns at **60.09%**; Premium on annual at **14.22%**. Critically, an annual Basic subscriber (26.91%) retains better than a monthly Premium one (36.82%) — so the intervention is not to fix the entry product, it is to change the terms the entry product is sold on.

3. Support contact is a timestamped early-warning signal.

Churn runs at **29.09%** among subscribers with no valid pre-outcome support contact, **43.59%** once a ticket is filed, and **67.77%** when that ticket escalates. The signal arrives while the customer is still a customer — and 45.25% of support-contacting customers reach an escalation, indicating a resolution problem feeding directly into the highest-risk cohort.

4. Risk is front-loaded in the first year.

58.5% of all churn occurs within twelve months of signup. Churned subscribers have a median tenure of 275 days against 570 for retained. Retention spend applied evenly across the lifecycle is being spent largely after the decision has been made.

The central recommendation

Pilot a migration of month-to-month subscribers onto annual terms, prioritising the Basic-monthly cohort at 60.09% churn, and route first support contact into a retention queue rather than waiting for escalation. Both actions rely on attributes the business already captures; neither requires new instrumentation. The contract action should be run as a randomised pilot, for the reason given in Section 06.

02 Contents

01	Executive Summary	2
02	Contents	3
03	Business Context & Objectives	4
04	Methodology & Data Preparation	5
05	Performance Baseline	6
06	Where Churn Concentrates	7
07	The Compounding Risk Profile	9
08	Support Contact as an Early-Warning Signal	10
09	Lifecycle & Timing of Churn	11
10	Geographic Exposure	12
11	Revenue & Lifetime Value Impact	13
12	Strategic Recommendations	14
13	Target State & Conclusion	16
A	Appendix A — Methodology Notes	17
B	Appendix B — Metric Definitions	19

How to read this report

Sections 03-05 establish the business problem, the method, and the aggregate baseline. Sections 06-11 are the analysis proper, each opening with the evidence and closing with what it implies. Section 12 converts those findings into seven ranked actions, each carrying the revenue or customer volume it addresses and an effort estimate. Section 13 and the appendices cover the target state, methodological decisions, and metric definitions.

Every figure in this report is built from the same cleaned dataset the accompanying Jupyter notebook produces, and every reported figure has been cross-checked against the source data. They present different views deliberately: the notebook carries the exploratory work — the monthly churn trend, churn by plan, and the correlation heatmap with its encoding-order comparison — none of which appear here, while this report adds the risk-tier, channel, support, geographic and revenue-exposure views a stakeholder audience needs.

03 Business Context & Objectives

A growing subscription streaming business is losing roughly a third of its customers and has no segmented view of who they are.

THE PROBLEM

Retention budget is currently applied uniformly across the subscriber base. That is expensive and largely wasted: most customers were never going to leave, and the ones who were are not being identified in time. The commercial question is not "how do we reduce churn" in the abstract — it is **which customers to contact, in what order, and on what trigger**.

OBJECTIVES

- Establish a churn and revenue baseline for the subscriber book
- Identify which segments carry disproportionate risk, and quantify their revenue weight
- Determine whether any behavioural signal precedes cancellation and can serve as an intervention trigger
- Locate where in the customer lifecycle risk is concentrated
- Deliver a prioritised, operationally actionable retention plan

BUSINESS CONTEXT

The company is growing. Annual signups rose from 5,664 in 2019 to 27,638 in 2025, so the subscriber base skews toward recent joiners and cumulative churn varies with cohort exposure. This is accounted for explicitly throughout rather than left to distort the findings — see Appendix A.

DATA ASSETS

Three related tables in a SQLite database, joined on customer ID. 100,000 subscribers across 51 US markets, observed January 2019 to December 2025.

Table	Records	Contents	Role in the analysis
customer	100,000	Name, date of birth, gender, state, country	Demographic and geographic segmentation
subscription	100,000	Plan, contract, start / renewal / cancellation dates, monthly charge, lifetime value, risk score	Churn outcome, revenue, tenure and segment attributes
support	18,718	Complaint date, escalation status, satisfaction score	Behavioural early-warning signal

The support table holds multiple rows per customer, which is the source of the join fan-out addressed in Section 04.

04 Methodology & Data Preparation

The analysis was built as a reproducible pipeline: SQL extraction, cleaning, feature engineering, then measurement.

1 • EXTRACTION

Tables were read directly from SQLite into pandas, with schema inspected via `PRAGMA table_info` before loading.

2 • CLEANING

- **Type correction:** five date fields stored as text converted to datetime.
- **Category standardisation:** gender values consolidated (Men→Male, Women→Female); a spelling inconsistency in the acquisition channel field ("Refferal") corrected, which would otherwise have split that category across every aggregation.
- **Missing-value recovery:** 1,728 absent country values recovered by deriving a state-to-country lookup from complete records rather than discarding the rows.
- **Structural nulls preserved:** absent cancellation dates indicate active subscribers; absent support fields indicate customers with no valid pre-outcome support contact. Both are information and were retained as such.

3 • JOIN INTEGRITY

Joining the three tables initially returned **102,551 rows from 100,000 customers**. The support table held 18,718 tickets belonging to only 16,167 customers, so joining on customer ID caused one-to-many matches to fan out and duplicate subscription records. Left unaddressed this would have inflated every revenue and churn figure in this report. The defect raises no error — validating row counts against the known customer count is what catches it.

Support tickets were first filtered to **pre-outcome** signals — for churned customers, only tickets dated before the cancellation date; for active customers, only tickets up to the fixed as-of date — retaining **16,002 of 18,718** tickets, since a ticket logged on or after cancellation cannot be an early warning. The surviving tickets were then aggregated to one row per customer: a complaint-count feature preserves repeat-contact information, and each customer's most recent pre-outcome ticket is retained, on the basis that the latest interaction best represents current escalation status and satisfaction. The corrected join returns exactly 100,000 rows, and **14,096** customers carry a pre-outcome support record.

4 • DERIVED FEATURES

Feature	Definition	Purpose
Churn flag	1 where a cancellation date exists	Binary outcome
Complaint count	Pre-outcome tickets filed per customer	Repeat-contact intensity
Escalation flag	Numeric encoding of escalation status	Correlation analysis
Tenure (days)	Start to cancellation, or to fixed as-of date if active	Lifecycle measurement
Support state	No valid pre-outcome contact / ticket filed / ticket escalated	Behavioural segmentation
Risk tier	Risk score banded Low / Medium / High	Outreach prioritisation

Tenure for active subscribers is measured to a fixed as-of date of 31 December 2025 — the close of the observation window — rather than to the current date, so every figure in this report reproduces identically on any future run.

05 Performance Baseline

The aggregate position of the subscriber book before segmentation.

Metric	Value	Interpretation
Subscribers	100,000	67,324 active · 32,676 cancelled
Churn rate	32.68%	Cumulative across a base with varying tenure
Retention rate	67.32%	Roughly two in three subscribers retained
Total monthly charges	\$1,494,659	\$1.06M from active subscribers
ARPU	\$14.95	Average monthly charge per subscriber record
Monthly charges lost to churn	\$433,107	28.98% of total monthly charges
Total lifetime value	\$31.18M	Average \$311.82 per subscriber
Lifetime value lost	\$6.12M	19.64% of accumulated book value
Average tenure	642 days	Retained 745 · churned 430
Support tickets	18,718	From 16,167 subscribers · 16,002 pre-outcome (0.16 each)
Escalation rate	45.25%	Of support-contacting customers (6.38% of all subscribers)

Reading the escalation rate

Escalation is reported two ways because they answer different questions. Of the **14,096** subscribers who contacted support before their outcome, **45.25%** had their most recent ticket escalated — while across the whole base only **6.38%** of customers reached an escalation. The first figure is the operational one, and it points at first-contact resolution as a process weakness independent of churn.

Risk distribution

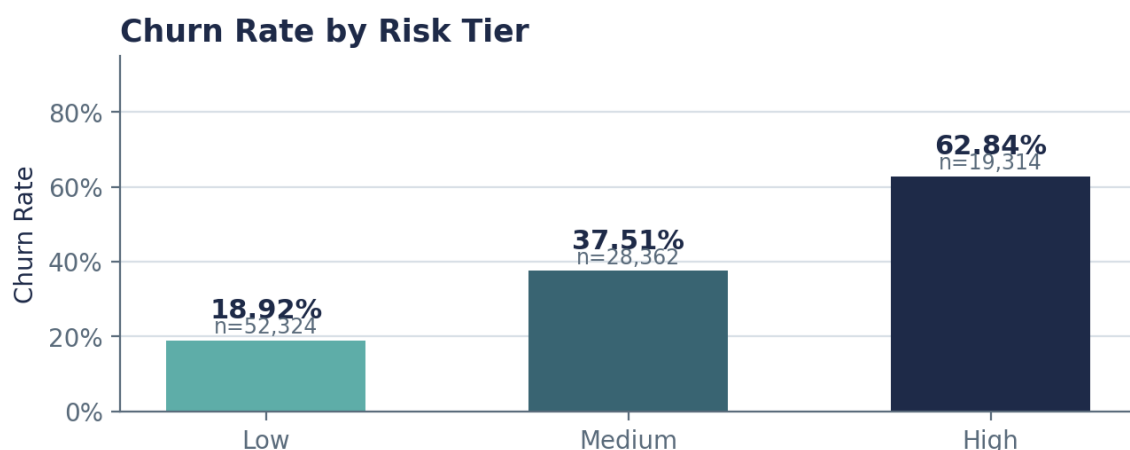


Figure 1: Observed churn by supplied risk tier. Tiers separate meaningfully without being deterministic.

The supplied risk score bands the base into Low (52,324 subscribers, 18.92% churn), Medium (28,362, 37.51%) and High (19,314, 62.84%). A high-risk subscriber is over three times more likely to cancel than a low-risk one — informative enough to sequence outreach, but not deterministic, which is what a usable score should look like. No finding or recommendation in this report depends on it; see Appendix A.

06 Where Churn Concentrates

Two attributes recorded at the point of sale separate churning subscribers from retained ones.

6.1 Contract type — the primary point of separation

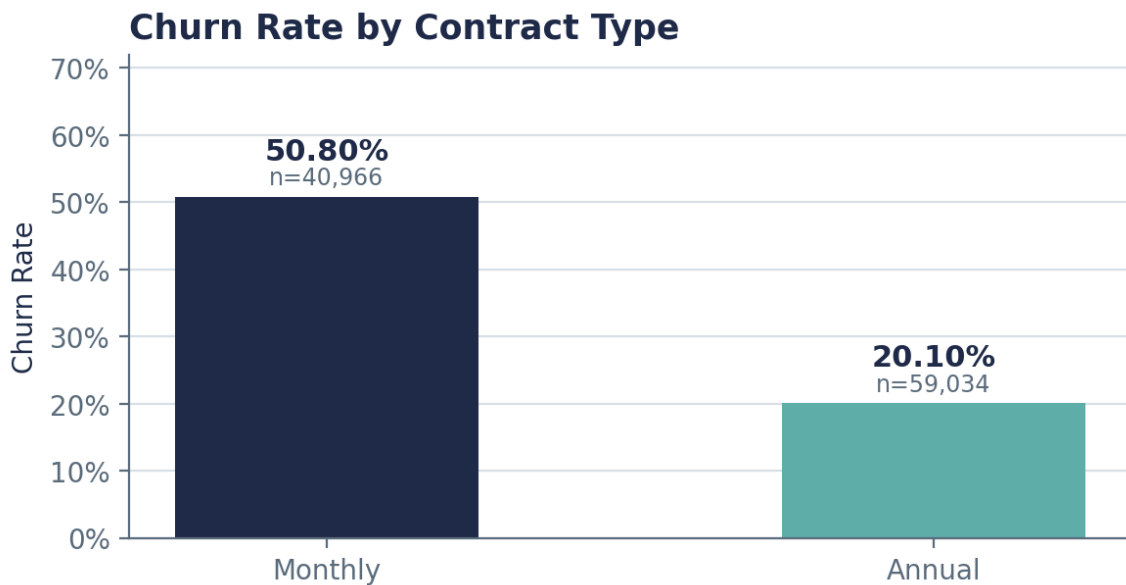


Figure 2: Churn rate by contract type. Monthly contracts churn at 2.5× the annual rate.

Monthly-contract subscribers churn at **50.80%** against **20.10%** for annual — a 2.5× difference across 40,966 and 59,034 subscribers.

Contract	Subscribers	Churn rate	MRR	MRR lost	Share of all loss
Monthly	40,966	50.80%	\$573,989	\$269,154	62.1%
Annual	59,034	20.10%	\$920,669	\$163,952	37.9%
Total	100,000	32.68%	\$1,494,659	\$433,107	100%

Monthly contracts hold **38.4% of total monthly charges** but **account for 62.1% of everything lost**. Nearly half of all monthly-contract revenue has already walked out the door.

6.2 What this comparison can and cannot establish

The 2.5× gap is an upper bound on what contract migration can deliver

Contract type is **chosen by the customer, not assigned by the analyst**. Two mechanisms inflate the observed gap beyond any effect the contract itself has:

Selection. Subscribers willing to commit to twelve months are, on average, already more certain they want the service. Some of the difference between 50.80% and 20.10% is the disposition of the people who choose each term, not the term acting on them.

Mechanical censoring. An annual subscriber largely cannot cancel mid-term. Part of the lower annual figure is definitional rather than behavioural, and migrating a flight-risk customer onto an annual contract may defer a cancellation to the renewal date rather than prevent it.

Both push in the same direction, so **30.7 percentage points is a ceiling, not an estimate**. The correct response is not to discard the finding — contract type remains the strongest separator in the dataset, and it is one of the few attributes the business actually controls — but to treat migration as a

hypothesis to be tested, not a settled result. Recommendation 1 is therefore specified as a randomised pilot, and the pilot's measured lift, not this cross-sectional gap, is what should size the opportunity.

This caution applies specifically to the causal reading. The descriptive finding is unaffected and remains directly usable: monthly-contract subscribers are where the losses are concentrated, they can be identified from data captured at signup, and they are the right population to target first regardless of how much of the gap turns out to be causal.

6.3 Acquisition channel

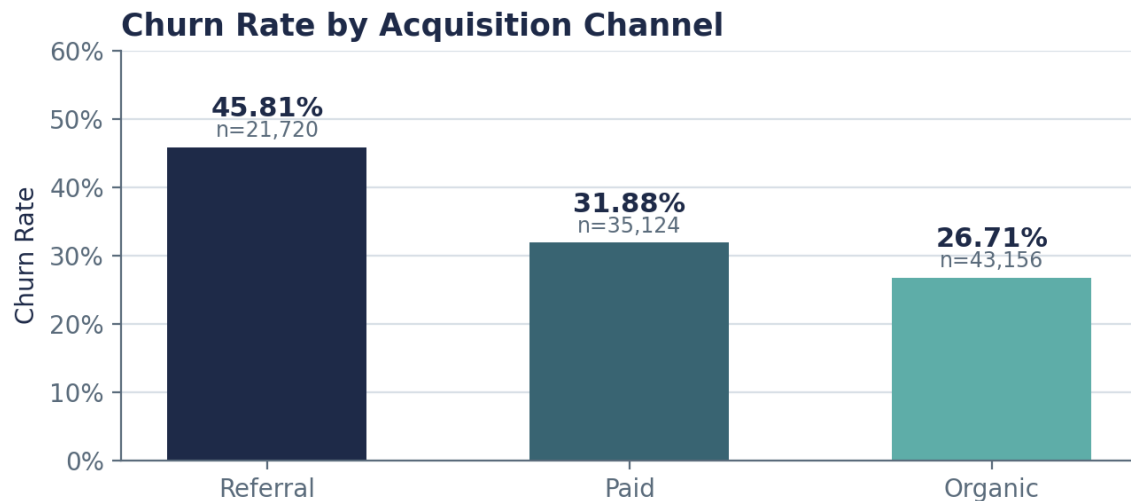


Figure 3: Churn rate by acquisition channel. Referral produces the highest churn and the lowest revenue.

Referral-acquired subscribers churn at **45.81%**, against 31.88% for paid and 26.71% for organic. This inverts a common assumption: referral is typically treated as a high-quality, low-cost channel, but here it produces the least durable customers and the smallest revenue base at \$324,554 — roughly half the organic channel's \$645,236.

Channel	Subscribers	Churn rate	MRR	Assessment
Referral	21,720	45.81%	\$324,554	Underperforming
Paid	35,124	31.88%	\$524,869	Average
Organic	43,156	26.71%	\$645,236	Strongest

The referral programme appears optimised for acquisition volume rather than retained value — a distinction worth resolving before further channel investment. As with contract type, channel is not randomly assigned, so this is a statement about which customers the channel brings in, not proof that the channel damages them.

07 The Compounding Risk Profile

Plan tier and contract type do not act independently. They compound — and contract dominates.

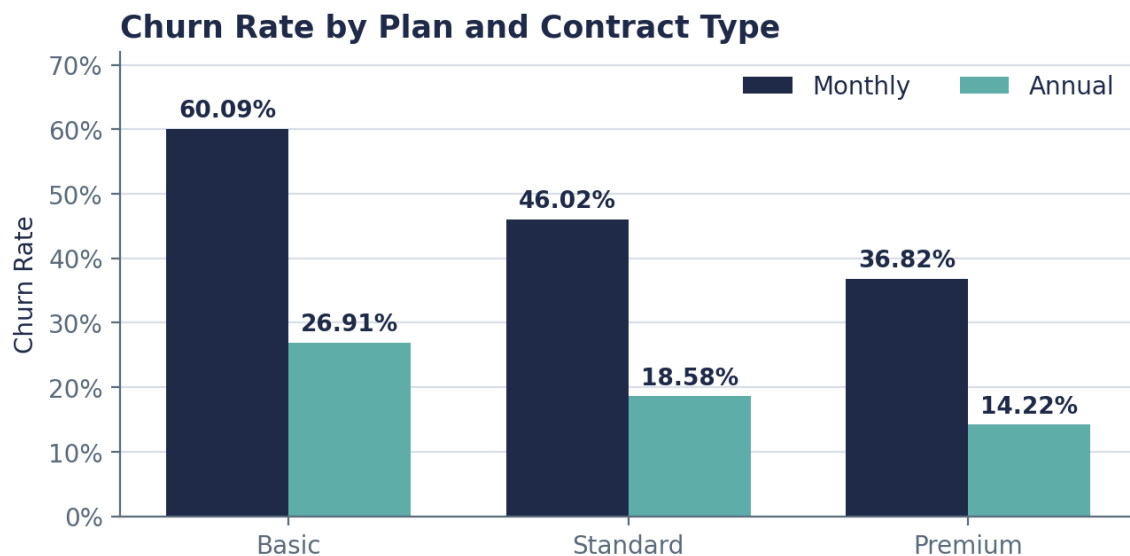


Figure 4: Churn rate across plan tier and contract type. Every tier churns far worse on monthly terms.

Plan	Monthly	Annual	Gap	ARPU	Avg CLTV
Basic	60.09%	26.91%	33.2 pts	\$9.72	\$171
Standard	46.02%	18.58%	27.4 pts	\$15.59	\$325
Premium	36.82%	14.22%	22.6 pts	\$22.63	\$527

Churn falls consistently as plan value rises, and falls again within every tier when the contract is annual. The worst cell in the matrix — **Basic on a monthly contract at 60.09%** — churns more than four times the best, Premium on annual at 14.22%.

The diagonal is the finding

An annual Basic subscriber churns at **26.91%**. A monthly Premium subscriber churns at **36.82%**. The cheaper product on a longer commitment retains better than the premium product on a short one.

This reframes the intervention. The instinct on seeing Basic at 42.69% overall is to fix the entry product — more content, better onboarding, repricing. The matrix says the higher-return move is to change the terms the entry product is sold on. Contract structure is also faster and cheaper to change than a product roadmap.

The ordering here is more robust than the size of any individual gap. Even if selection explains a substantial share of the contract effect, the fact that it dominates plan tier within every price point is difficult to attribute to price sensitivity alone, and it survives as a planning assumption while the pilot in Recommendation 1 measures the true magnitude.

08 Support Contact as an Early-Warning Signal

Segment attributes identify who is at risk. Support activity identifies when.

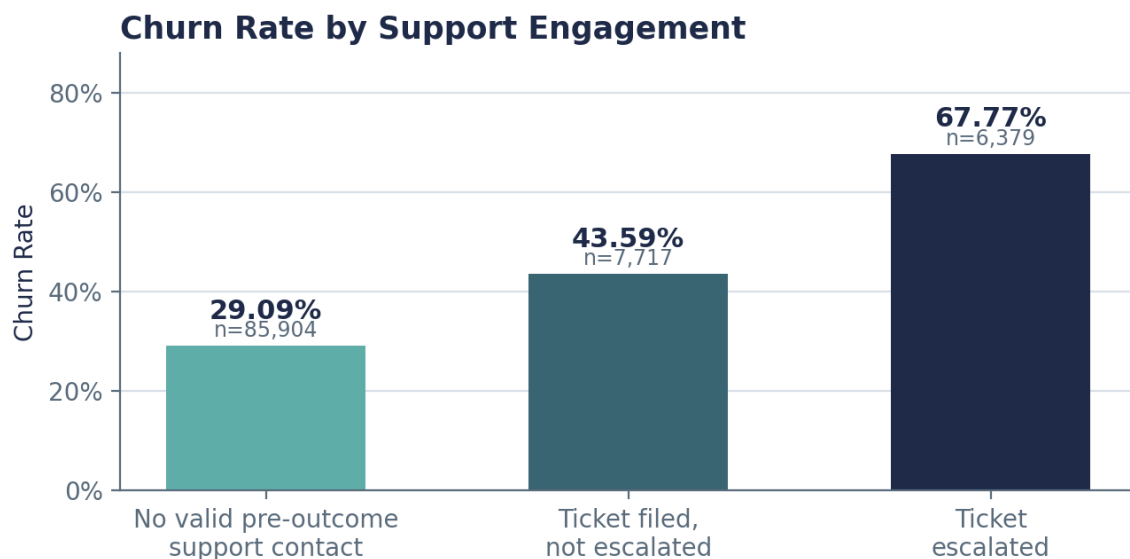


Figure 5: Churn rate by support engagement level. Risk rises monotonically with escalation.

Support state	Subscribers	Churn rate	Lift vs baseline
No valid pre-outcome contact	85,904	29.09%	—
Ticket filed, not escalated	7,717	43.59%	+14.5 pts
Ticket escalated	6,379	67.77%	+38.7 pts

A subscriber who files a ticket is 1.5× more likely to cancel than one with no valid pre-outcome support contact; one whose ticket escalates is 2.3× more likely.

WHY THIS MATTERS OPERATIONALLY

Segment attributes are fixed — a subscriber's acquisition channel cannot be changed after signup. Support contact is an event: timestamped, already logged, and occurring while the customer is still a customer. It is the only signal in this dataset that provides both a warning and a window in which to act.

The escalation rate compounds the problem. **45.25% of support-contacting customers reach escalation** on their most recent ticket — an operational weakness that is actively manufacturing the highest-risk cohort in the book.

The ladder understates the signal rather than inflating it

Counting only pre-outcome tickets means churned subscribers are observed for a shorter window than active ones — 430 days against 745 — so they have less opportunity to file a ticket at all. Adjusted for that exposure, churned subscribers file **0.65 pre-outcome tickets per 1,000 customer-days against 0.14** for retained, a **4.7× difference**. The ladder is a conservative reading, not an optimistic one (Appendix A).

STATED CANCELLATION REASONS

Recorded reasons reinforce the picture: **Too expensive (23.7%)** and **Switched to competitor (21.0%)** lead, followed by Not enough content (16.6%), Poor streaming quality (11.2%) and Technical issues (9.3%). Pricing and competitive pressure account for 44.6% of cancellations; product and technical quality a further 37.2% — all issues that surface through a support channel before they surface as a cancellation.

09 Lifecycle & Timing of Churn

Risk is not distributed evenly across the customer lifecycle — it is heavily front-loaded.

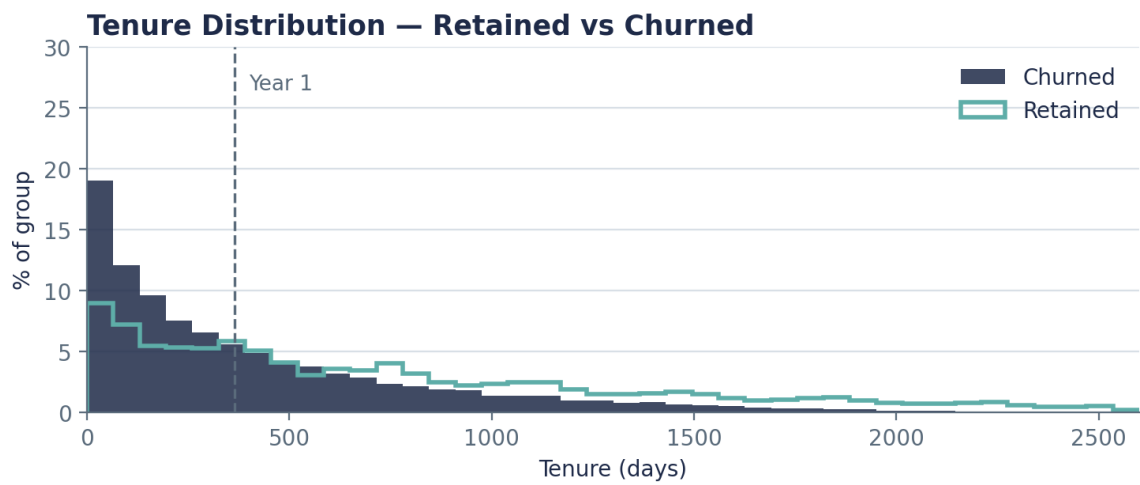


Figure 6: Tenure distribution, each cohort shown as a share of itself. The retained group is roughly twice the size, so raw counts would not compare fairly; the retained series is drawn as an outline so the two are not read as stacked.

Tenure band	Churned	Retained	Reading
Under 180 days	38.6%	20.7%	Churn concentrates immediately after signup
Over 4 years	4.1%	15.6%	Long-tenure subscribers rarely leave
Median tenure	275 days	570 days	Churners last roughly half as long

58.5% of all churn occurs within the first twelve months. This is a classic front-loaded hazard curve: risk peaks immediately after signup and declines steadily as subscribers establish a viewing habit. Subscribers who pass the two-year mark are markedly unlikely to cancel.

Implication for retention spend

Retention budget applied evenly across the lifecycle is being spent largely on subscribers who have already decided to stay. The evidence locates the decisive window precisely: onboarding and the first twelve months, where more than half of all attrition originates. Reallocating spend toward that window costs nothing extra — it is a sequencing change, not a budget increase.

A NOTE ON COHORT EXPOSURE

The headline 32.68% is cumulative churn across a base with varying tenure — a 2019 joiner has had seven years in which to cancel, a 2025 joiner only months. This is why churn by signup year appears to decline over time: it reflects exposure, not improving retention. Comparing subscribers with equal exposure, **twelve-month churn is stable at 21-22% across every cohort** (21.3% for 2019 through 22.0% for 2024), confirming the underlying hazard has not changed. That twelve-month rate — not the cumulative figure — is the metric the target state in Section 13 is set against.

10 Geographic Exposure

Churn varies materially by market, and the pattern is structural rather than random.

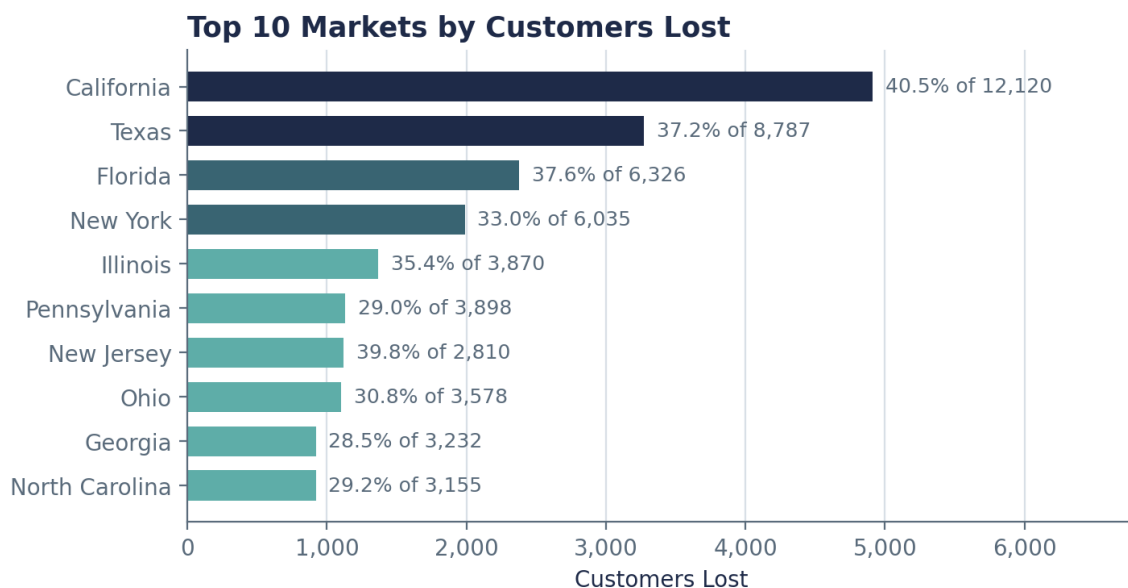


Figure 7: The ten markets losing the most subscribers in absolute terms, with each market's churn rate and base size.

Rates range from **15.52% in Vermont to 41.90% in the District of Columbia**. The highest-churn markets are consistently the most urbanised — DC, California (40.54%), New Jersey (39.82%), Arizona (39.47%), Florida (37.62%) — while the lowest are rural: Vermont, Maine (19.02%), Montana (19.52%).

Market	Subscribers	Churn rate	Customers lost	MRR lost
California	12,120	40.54%	4,914	\$65,988
Texas	8,787	37.24%	3,272	\$43,191
Florida	6,326	37.62%	2,380	\$31,762
New York	6,035	32.96%	1,989	\$26,141
Illinois	3,870	35.37%	1,369	\$18,126
Pennsylvania	3,898	29.04%	1,132	\$14,871
Top six	41,036	36.69%	15,056	\$200,079

Rate alone is the wrong priority ranking

The District of Columbia has the highest churn rate in the book at 41.90% — across **210 subscribers**. California churns at 40.54% across 12,120, losing 4,914 customers and \$65,988 in monthly revenue. Ranked by rate, DC leads. Ranked by customers and revenue actually lost, California is more than twenty times the priority. Geographic intervention should be sequenced by **rate multiplied by volume**, which is why base sizes are printed against every market in this report.

Sampling error reinforces the point. On 210 subscribers, DC's 41.90% carries a 95% confidence interval of roughly **±6.7 points (35.2%–48.6%)** — wide enough to overlap several markets ranked below it. California's 40.54%, on 12,120 subscribers, is precise to **±0.9 points**. The difference between the two headline rates is well inside the noise on the smaller market; the difference in absolute loss is not.

The six markets above account for **46.1% of all customers lost** while holding 41.0% of the base — a concentration that makes a targeted regional programme viable where a uniform national campaign would not be.

11 Revenue & Lifetime Value Impact

Churn has removed 28.98% of the book's monthly charges — and the loss is structurally concentrated.

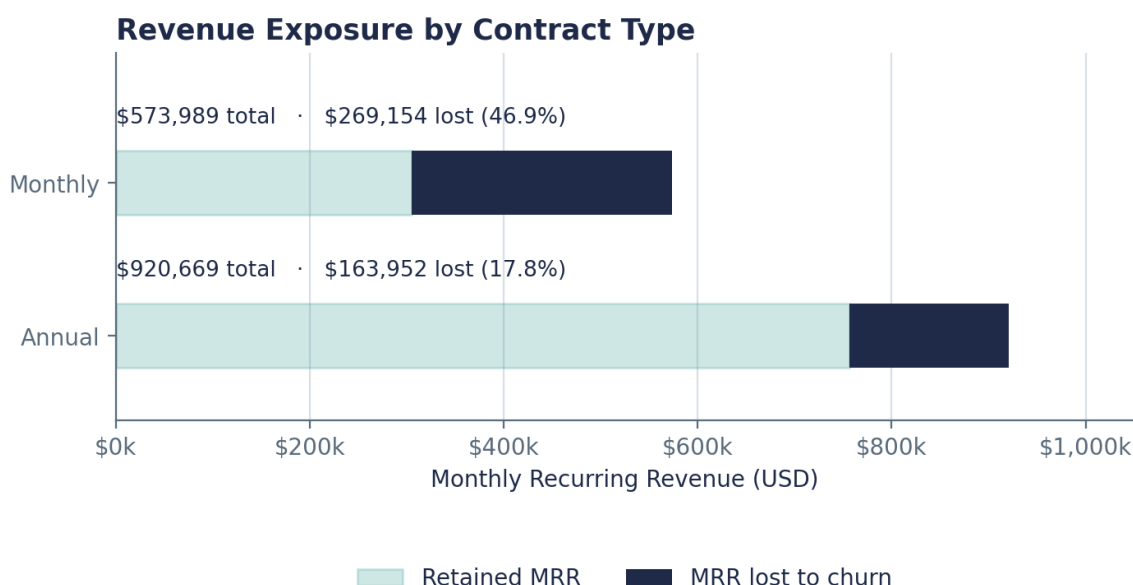


Figure 8: Monthly charges by contract type, showing the portion already lost to churn.

Cancelled subscribers represented **\$433,107** of the book's **\$1,494,659** in total monthly charges. The loss is heavily skewed by contract type: monthly contracts hold 38.4% of monthly charges but account for 62.1% of it, and **46.9% of all monthly-contract revenue is already gone**.

11.1 Lifetime value

Cohort	Subscribers	Avg CLTV	Total CLTV
Retained	67,324	\$372	\$25.06M
Churned	32,676	\$187	\$6.12M
Total	100,000	\$312	\$31.18M

Churned subscribers carry roughly half the lifetime value of retained ones — \$187 against \$372, a correlation with churn of **-0.28**. The business is losing its lower-value customers rather than its best accounts, which reframes this as a low-end retention and conversion problem rather than an erosion of the premium base.

11.2 Where the revenue sits

Segment	MRR	MRR lost	% of segment lost
Standard plan	\$657,931	\$194,151	29.5%
Premium plan	\$481,735	\$91,615	19.0%
Basic plan	\$354,992	\$147,340	41.5%
Total	\$1,494,659	\$433,107	28.98%

Ranked by rate, Basic is worst at 41.5% of its revenue lost. Ranked by absolute revenue lost, Standard is largest at \$194,151 — it holds the biggest customer base and the biggest revenue block. An effective retention programme addresses both: **Basic for severity, Standard for scale**.

12 Strategic Recommendations

Seven actions, ordered by expected impact against implementation effort.

CRITICAL

1. Pilot migration of month-to-month subscribers onto annual terms

Monthly contracts churn at 50.80% against 20.10% on annual, and account for 62.1% of all revenue lost while holding 38.4% of monthly charges. Prioritise the Basic-monthly cohort, which churns at 60.09%.

Run the conversion offer as a **randomised pilot** — a treated group receiving the annual-commitment discount against a held-out control drawn from the same monthly population. As set out in Section 06, the observed 30.7-point gap is inflated by self-selection and by the fact that annual subscribers cannot easily cancel mid-term, so it is a ceiling rather than an expected effect.

Size the break-even discount from the pilot's **measured retention lift, contribution margin, and expected remaining customer lifetime** — not by comparing the discount against the cumulative churn rate, which measures a different quantity over a different horizon and would substantially overstate the headroom. Track whether the pilot prevents cancellations or merely defers them to first renewal.

Addresses: \$269,154 of \$433,107 in monthly charges lost · **Effort:** Medium — offer design, control-group setup and campaign economics

CRITICAL

2. Default referral signups to annual contracts

Referral-acquired subscribers churn at 45.81% while generating the lowest revenue of the three channels at \$324,554. The programme is currently acquiring the least durable customers in the book.

Make annual the default term offered to referred customers and tie the referral incentive to annual signup rather than to acquisition alone. Where monthly is retained as an option, route those subscribers into a structured first-90-day onboarding sequence. Because this changes a default rather than a price, it can be A/B tested at low cost and will independently help validate the contract hypothesis in Recommendation 1.

Addresses: 21,720 subscribers in the highest-churn channel · **Effort:** Low — a pricing and offer configuration change

HIGH

3. Trigger retention outreach at first support contact, not at escalation

Churn rises from 29.09% to 43.59% the moment a ticket is filed, and reaches 67.77% on escalation.

Waiting for escalation means waiting for the situation to deteriorate — and 45.25% of support-contacting customers do reach escalation. Adjusted for exposure, churned subscribers file tickets at 4.7× the rate of retained ones, so the signal is stronger than the headline ladder suggests.

Route every new ticket from a subscriber carrying elevated risk attributes, or with above-median lifetime value, into a retention queue running in parallel with normal support handling.

Addresses: 14,096 subscribers with an active behavioural signal · **Effort:** Medium — support workflow integration

HIGH

4. Concentrate retention spend in the first twelve months

58.5% of all churn occurs within a subscriber's first year, and churned subscribers have a median tenure of just 275 days. Retention effort spread evenly across the lifecycle is largely arriving after the decision has been made.

Reweight onboarding, early engagement prompts and first-renewal offers toward months 1–12. This is a sequencing change rather than a budget increase.

Addresses: the window containing more than half of all attrition · **Effort:** Low — reallocation of existing spend

HIGH

5. Audit first-contact resolution in support

45.25% of support-contacting customers reach escalation. Independent of churn, that means nearly half of support contacts fail to resolve at first touch — and escalated tickets precede a 67.77% churn rate.

Audit escalation causes across the ticket base to establish whether the driver is product defect, policy friction, or limits on agent authority. Track escalation rate as a standing retention metric, not solely a support-quality metric.

Addresses: 6,379 subscribers in the highest-risk state in the book · **Effort:** Medium — process review

MEDIUM

6. Sequence geographic intervention by volume, not by rate

California loses 4,914 subscribers and \$65,988 in monthly revenue — more than twenty times the District of Columbia, which has a marginally higher churn rate on 210 subscribers and a confidence interval wide enough to overlap markets ranked well below it. Texas and Florida follow.

Competition intensity in dense urban markets argues for differentiated retention offers in the top handful of states rather than a uniform national campaign.

Addresses: the six markets containing 46.1% of all customers lost · **Effort:** Medium — regional offer design

MEDIUM

7. Reposition the Basic tier

Basic churns at 42.69% on the lowest ARPU (\$9.72) and lowest lifetime value (\$171). Its subscribers most frequently cite pricing and content depth as cancellation reasons.

Determine whether the tier under-delivers value or attracts low-intent customers from the outset. Options include repositioning it explicitly as a trial step with a defined upgrade path into Standard, or adjusting its content allocation. Sequence this after Recommendation 1 reports: if contract terms carry most of the effect, a product rebuild is the more expensive route to the same outcome.

Addresses: \$147,340 of monthly charges lost · **Effort:** High — product and pricing decision

13 Target State & Conclusion

Churn in this book is concentrated, attributable, and — unusually — visible in advance.

Two attributes recorded at the point of sale separate most of it. Contract type distinguishes a 50.80% cohort from a 20.10% one; acquisition channel separates 45.81% from 26.71%. With plan tier they compound into a matrix where the worst cell churns at 60.09% and the best at 14.22% — none of it requiring a predictive model, since the attributes are already captured at signup. Layered on top, support contact provides a timestamped behavioural trigger, with churn rising by approximately 50% once a ticket is filed and more than doubling when the latest ticket is escalated. That gives the business two complementary instruments: **a static profile identifying who to watch from day one, and an event identifying when to intervene.**

Proposed target state

Targets are set on twelve-month cohort churn, not the cumulative rate

The headline 32.68% is cumulative churn to date — a stock measure that can only rise as the book ages, since every customer who has already cancelled stays cancelled. It cannot serve as a target. The measures below use **twelve-month churn**: of subscribers with at least a year of exposure, the share who cancelled within 365 days. That is a flow measure, comparable across cohorts, and it moves when retention improves. Baselines cover the 72,362 subscribers with a full year of exposure. Targets are directional: the first four depend on the size of the causal contract effect that Recommendation 1's pilot is designed to measure, and should be revised once it reports.

Measure (12-month cohort basis)	Current	Target	Primary mechanism
Overall 12-month churn	21.8%	< 16%	Contract migration
Monthly-contract 12-month churn	36.8%	< 26%	Annual conversion pilot
Referral-channel 12-month churn	34.0%	< 24%	Annual-default onboarding
Basic-monthly 12-month churn	46.6%	< 32%	Combined contract + tier action
Monthly contracts as share of new signups	41.0%	< 28%	Contract mix shift
Ticket escalation rate	45.3%	< 25%	First-contact resolution

Sequencing

- **Immediate:** change the default contract term offered to referral customers. Lowest effort, addresses the highest-churn channel, needs only a configuration change — and gives an early read on the contract hypothesis.
- **Next 90 days:** launch the randomised annual-conversion pilot to existing monthly subscribers, prioritised by plan tier and risk score, with a held-out control. Instrument support tickets to flag retention risk at first contact, not at escalation.
- **Next two quarters:** read out the pilot and revise the targets above; complete the escalation-cause audit; reweight retention spend toward months 1–12; decide the position of the Basic tier; stand up differentiated offers in the six highest-loss markets.

Closing assessment

The retention problem is structural, not diffuse: short-commitment contracts sold into a low-intent channel produce subscribers who leave within their first year, and the business already generates every signal needed to spot them. The highest-return intervention is also the simplest — change the terms at-risk subscribers are acquired on, and act on support contact the first time it occurs. What remains open is how much of the contract gap is the contract and how much is the customer; one pilot answers that in a quarter.

A Appendix A — Methodology Notes

Decisions taken during the analysis and their effect on how the findings should be read.

CUMULATIVE CHURN, NOT AN ANNUAL RATE

The headline 32.68% is cumulative across a base with varying tenure. A 2019 joiner has had seven years in which to cancel; a 2025 joiner only months. Churn by signup year therefore appears to decline over time, reflecting exposure rather than improving retention. Comparing subscribers with equal exposure, twelve-month churn is stable at 21–22% across every cohort — the underlying hazard has not changed. Section 13 sets its targets on that twelve-month measure for this reason.

OBSERVATIONAL DATA, NOT AN EXPERIMENT

Every segment in this analysis is self-selected: customers choose their contract term, plan tier and, in effect, their acquisition channel. Differences between segments therefore combine any effect of the attribute with the disposition of the people who choose it, and in the case of contract type an annual subscriber is additionally constrained from cancelling mid-term. The observed gaps are consequently **upper bounds on what intervention can deliver**, and Recommendation 1 is specified as a randomised pilot for exactly this reason. The descriptive finding — that losses concentrate in identifiable, pre-observable segments — does not depend on the causal question and stands regardless.

REPRODUCIBILITY

Tenure for active subscribers is measured to a fixed as-of date of 31 December 2025, the close of the observation window, rather than to the date of execution. Measuring to "today" would cause reported tenure to drift upward on every subsequent run and silently contradict the written figures. Every number in this report reproduces exactly from the source database, and every figure regenerates from the cleaned dataset the notebook produces.

TREATMENT OF THE SUPPLIED RISK SCORE

The source data includes a pre-computed churn score. Its tiers separate meaningfully without being deterministic — 18.92%, 37.51% and 62.84% observed churn across Low, Medium and High — and it is used here to sequence outreach and demonstrate segmentation method. It is also strongly associated with contract type (–0.68), which suggests it is partly derived from the same attributes analysed independently elsewhere in this report; treating it as a separate driver would double-count. **No finding or recommendation depends on it.** Every conclusion rests on independently observable attributes: contract type, plan tier, acquisition channel, support activity, tenure, geography and revenue.

PRE-OUTCOME FILTERING AND EXPOSURE ASYMMETRY

Support tickets logged on or after a cancellation cannot serve as an early warning, so they are excluded — 16,002 of 18,718 tickets survive, covering 14,096 customers. This creates an asymmetry: churned subscribers are observed for an average of 430 days against 745 for retained ones, giving them less opportunity to file a ticket at all. The bias runs against the finding. On an exposure-adjusted basis churned subscribers file 0.65 pre-outcome tickets per 1,000 customer-days against 0.14 for retained — a 4.7× difference — so the support ladder is a conservative statement of the relationship.

COMPARING GROUPS OF UNEQUAL SIZE

The tenure distribution is plotted as a percentage of each cohort rather than as raw counts, because the retained group is roughly twice the size of the churned one. On raw counts the two shapes are not comparable and the separation between them is not visible. The retained series is drawn as an outline rather than a second filled series so the overlap cannot be misread as a stacked total.

CORRELATION ON BINARY VARIABLES

The escalation–churn correlation of 0.20 understates the relationship, as Pearson correlation between two binary variables is bounded by their base rates. The support ladder (29.09% → 43.59% → 67.77%) is a more faithful presentation of the same finding and is what this report leads with.

SMALL-SAMPLE MARKETS

State-level rates are computed on bases ranging from 174 subscribers (Vermont) to 12,120 (California). Rates on the smallest markets carry confidence intervals of $\pm 6-7$ points and should not be ranked against each other as though they were precise; DC's 41.90% on 210 subscribers is statistically indistinguishable from several markets below it. Priority is therefore set by absolute customers and revenue lost, where the largest markets are also the most precisely measured.

JOIN INTEGRITY

The initial three-table join produced 102,551 rows from 100,000 subscribers, caused by a one-to-many fan-out from customers with multiple support tickets. This was detected by validating row counts against the known customer count and corrected by aggregating support to one row per customer before joining. Left unaddressed it would have inflated every revenue and churn figure in this report.

STRUCTURAL NULLS

Absent values were assessed for meaning before any treatment. Missing cancellation dates indicate active subscribers; missing support fields indicate customers with no valid pre-outcome support contact. Both were preserved as information. Only genuinely missing reference data — 1,728 country values — was imputed, using a state-to-country mapping derived from complete records.

B Appendix B — Metric Definitions

Metric	Definition	Value
Churn rate (cumulative)	Cancelled subscribers ÷ total subscribers, across the full observation window	32.68%
Retention rate	100% – cumulative churn rate	67.32%
12-month cohort churn	Share of subscribers with ≥365 days of exposure who cancelled within 365 days	21.8%
ARPU	Mean monthly charge per subscriber record	\$14.95
Total monthly charges	Sum of monthly charges across the book	\$1,494,659
Active MRR	Monthly charges from active subscribers	\$1,061,552
Monthly charges lost to churn	Monthly charges attributable to subscribers who have already cancelled	\$433,107
Share of charges lost	Monthly charges lost ÷ total monthly charges	28.98%
CLTV	Accumulated lifetime value per subscriber	\$311.82 avg
Tenure	Start to cancellation, or to 31 Dec 2025 if active	642 days avg
Complaints per user	Pre-outcome tickets ÷ total subscribers	0.16
Escalation rate (support-contacting)	Escalated most-recent ticket ÷ support-contacting customers	45.25%
Escalation rate (all customers)	Subscribers with an escalation ÷ all subscribers	6.38%
Support state	No valid pre-outcome contact / ticket filed / ticket escalated	3 levels
Risk tier	Supplied churn score banded <50 / 50-69 / 70+	Low / Med / High
Year-one churn share	Churners with tenure < 365 days ÷ all churners	58.5%
Retention curve (cover)	Kaplan-Meier estimate of the share still active at each tenure day	78.3% at year 1

SOURCE & REPRODUCIBILITY

All figures derive from a SQLite database of customer, subscription and support tables covering 100,000 subscribers across 51 US markets, January 2019 to December 2025. The cleaning, feature engineering and KPI derivation are documented in the accompanying Jupyter notebook, which produces the cleaned dataset this report is built from.

Prepared by Shayan Yawar Bhatti · Python, pandas, SQL · As-of date 31 December 2025.